The Media Connectivity Association



OVERVIEW AND UPDATES · AI IN MEDIA SERVICES

AI in Media Services

Evaluation framework for AI/ML models in media processing for 3GPP systems

3GPP

OCTOBER 2026 – www.5g-mag.com

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About the project

The reports the evaluation framework is built around

3GPP TR 26.847

Technical Report, V19.0.0

Evaluation of AI and ML in 5G media services: the evaluation testbed architectures and anchors, the metrics and the evaluation scenarios.

3GPP TR 26.927

Technical Report, V19.0.0

Study on AI and ML in 5G media services: the media use cases and the architectures for AI/ML model distribution and split operation.

3GPP TR 22.874

Technical Report, V18.2.0

Study on traffic characteristics and performance requirements for AI/ML model transfer in 5GS.

FIND OUT MORE **Technical Analysis** **Specifications** **Testbeds**

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The evaluation framework

What TR 26.847 defines for the evaluation of AI/ML in media services

Testbed architectures clause 4, common to every scenario unless stated

- Anchors: inference on the device, or in the network
- Split inference intermediate data testbed
- Model data testbed

Metrics clause 4.4, with the AI/ML task metrics

- AI/ML task metrics
- Video quality
- Complexity
- Bitrate
- Split model size
- Intermediate data size or bitrate
- Compressed model size
- Compressed intermediate data ratio
- Latency

Scenario template annex A, one entry per evaluation scenario

- Motivation and use case relevance
- Description
- Anchor model
- Testbed architecture and anchors
- Test configuration
- Metrics and requirements
- Test dataset and scripts
- Test conditions
- Interoperability

The functional side of AI/ML in media services is documented in TR 26.927; TR 26.847 is its evaluation companion.

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Scenarios

The four evaluation scenarios of TR 26.847 and the scenarios of the repository

In TR 26.847 the four evaluation scenarios of clause 5

- Compressed AI/ML model data for speech recognition
- Split inferencing for object detection and labelling
- Bit-incremental transmission and deployment of AI/ML models
- Real-time translation over IMS

Scripts in the repository read from the README and the scripts folders

- Compressed model transmission for speech recognition
- Split inferencing for object detection
- IMS real-time translation

Planned described in the README, with no scripts yet

- Split inferencing for human pose estimation
- Split inferencing for hand gesture recognition
- Video super resolution in video call
- Federated learning

The repository is an evaluation tool under development, not a service or a network function.

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What the scripts do

The three scenarios that have scripts today

COMPRESSED MODEL TRANSFER

Speech recognition models

A pre-trained model (wav2vec 2.0 or HuBERT) is compressed before it is sent to a device. A codec pipeline encodes and decodes it and reports its size and its recognition performance before and after.

SPLIT INFERENCING

Object detection

A pre-trained model detects objects in a video sequence and is split in two parts, one on the device and one in the network. Scripts convert and split a RetinaNet model and run it whole or split.

TRANSLATION

Translation models

Scripts train a Transformer translation model on Tatoeba sentence pairs and translate sentences, with split translation and a WebRTC translator. Four trained models are stored with Git LFS.

A Dockerfile builds the evaluation environment (Ubuntu 22.04, with Nvidia GPU support). The datasets are hosted outside the repository.

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How it grew

What each 3GPP release added to AI and ML in media services

Release 18

The SA1 study of the traffic of AI/ML model transfer, TR 22.874, is approved at version 18.0.0. Its Stage 1 requirements are in TS 22.261 (clause 6.40).

Release 19

The SA4 study FS_AI4Media: TR 26.927 and TR 26.847 are approved at SA#108 and created as version 19.0.0 (June 2025).

Release 20

Work plan of 28 September 2026. TR 26.927 recommends stage 3 work on AI/ML in IMS (TS 26.114 and TS 26.264) and TR 26.847 recommends a new evaluation study. These are recommendations, not decisions.

From the Standards pages of the website, and clause 7 of TR 26.927 and clause 6 of TR 26.847.

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Repositories

The repository of the AI/ML evaluation framework

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE
rt-ai-ml-evaluation-framework	the implementation of the AI/ML evaluation framework defined in 3GPP SA4 TR 26.847 (Release 19)	Compressed model transfer Split inferencing Translation scenario	On request Early Access

FIND OUT MORE

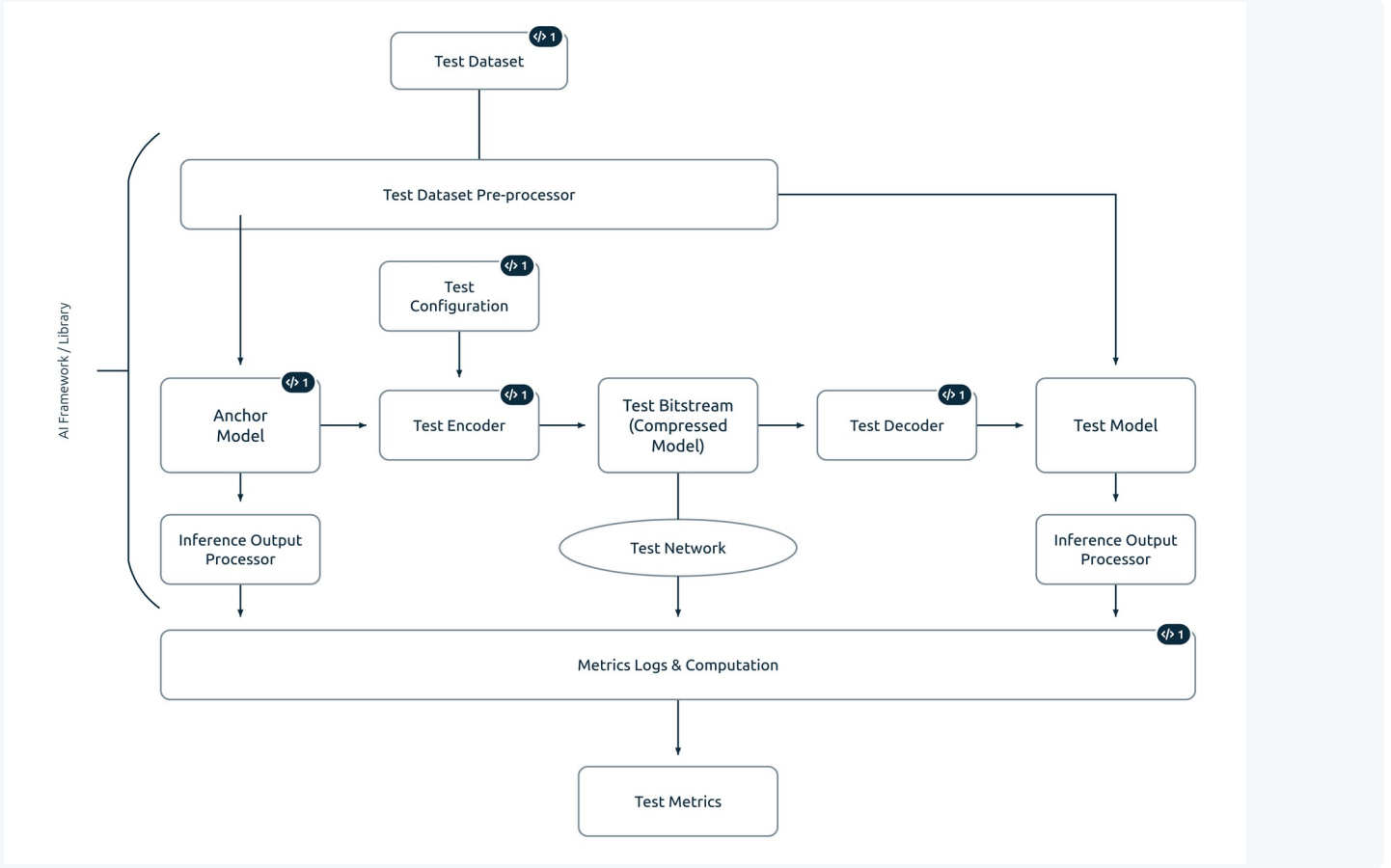
Technical Analysis | Specifications | Testbeds

● released ● no release yet

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Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

1 **rt-ai-ml-evaluation-framework**

Boxes without a badge are not mapped to a repository.

CONTRIBUTE

Latest updates and roadmap

Status on 22 October 2026

IN DEVELOPMENT

Bit incremental transfer

A pull request is open to merge the bit incremental transfer feature into the main branch.

NO SCRIPTS YET

Human pose estimation

The README describes the split inferencing of PoseNet at different layers; the repository has no scripts for it yet.

Video super resolution

The README gives the motivation: video calls are typically 480p today and XR calls need more. The repository has no scripts for it yet.

Hand gestures and federated learning

The README has the headings and no text. Both are planned.

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